

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B.Tech. Theory Examination (ME/CL/EE)****April-2018****MA241 Engineering Mathematics III****Date: 26/03/2018, (Monday)****Time: 01:30 pm to 04:30 pm****Maximum Marks: 70****Instructions:**

- i. The question paper contains two sections.
- ii. Section I and II must be attempted in separate answer sheets.
- iii. Figures to the right indicate marks.
- iv. Make suitable assumptions and draw neat figure where it is required.
- v. All notations and terminology are standard.

Section-I**Q-1 Choose the correct answer from the given options in the following: [07]**

I. Which one of the following is an even function? **01**
 (a) $\sin t$. (b) $\cos t$. (c) $\operatorname{cosec} t$. (d) $\tan t$.

II. The period of $\sin pt$ is **01**
 (a) $\frac{2\pi}{p}$. (b) 2π . (c) $\frac{\pi}{p}$. (d) $2\pi p$.

III. The $L[\sin 2t]$ is **01**
 (a) $\frac{4}{s^2 + 4}$. (b) $\frac{2}{s^2 + 4}$. (c) $\frac{4}{s^2 + 2}$. (d) $\frac{2}{s^2 + 2}$.

IV. $L^{-1}\left[\frac{1}{s}\right]$ is **01**
 (a) 1. (b) t . (c) t^2 . (d) 0.

V. If $L[f(t)] = F(s)$ then $L[e^{at} f(t)]$ is **01**
 (a) $F(s - a)$. (b) $F(s + a)$. (c) $F(t - a)$. (d) $F(t + a)$.

VI. Laplace transform of $f(t)$ is defined for **01**
 (a) Only for positive value of t . (b) Only for negative value of t .
 (c) Both positive and negative value of t . (d) Cannot say anything.

VII. The one dimensional heat equation is **01**
 (a) $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$. (b) $\frac{\partial u}{\partial t} = c^2 \frac{\partial u}{\partial x}$.
 (c) $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial u}{\partial x}$ (d) None of these.

Q-2 Attempt any Four. [16]

- I.** Obtain the Fourier series to represent the function $f(x) = x^2$ in $[0, 2\pi]$. **04**
- II.** Express $f(x) = e^x$ as a half – range cosine series for $0 < x < \pi$. **04**
- III.** Find the Fourier series for the following function $f(x) = \begin{cases} 0, & -2 < x < 0 \\ 1, & 0 < x < 2 \end{cases}$. **04**
- IV.** Using the Laplace transform solve $\frac{d^2 y}{dx^2} + y = 1$, where $y(0) = 1, y'(0) = 0$. **04**
- V.** Evaluate $L[t \cos at]$ and $L\left[\frac{\sin t}{t}\right]$. **04**
- VI.** A rod of 30cm long has its end A and B are kept at 20° and 80° respectively until steady state condition prevail. The temperature at each end is suddenly reduced to 0° and kept so. Find the resulting temperature $u(x, t)$ from the end A. **04**

Q-3(a) Attempt any Two. [06]

- I.** Evaluate $L^{-1}\left[\frac{2s^2 - 4}{(s+1)(s-2)(s-3)}\right]$. **03**
- II.** Obtain the Laplace transform of the function $f(t) = \begin{cases} 3, & 0 < t < 3 \\ 0, & t > 3 \end{cases}$. **03**
- III.** Using Convolution Theorem, Evaluate $L^{-1}\left[\frac{1}{s^2(s-1)}\right]$. **03**

Q-3(b) An impulsive voltage $E\delta(t)$ is applied to a circuit consisting of L, R, C in series with zero initial conditions. If i be the current at any subsequent time t , find the limit of i as $t \rightarrow 0$? **[06]**

OR

Q-3(b) A string of length $L = \pi$ has its ends fixed at $x = 0$ and $x = \pi$. At time $t = 0$ the string is given a shape defined by $f(x) = 50x(\pi - x)$ m, and then it is released. Find the deflection of the string at any time t . **[06]**

Section-II

Q-4 Choose the correct answer from the given options in the following: [07]

- I.** The Eigen values of a matrix $\begin{bmatrix} -7 & 1 \\ 0 & 0 \end{bmatrix}$ are **01**
 (a) $-7, 0$. (b) $-7, 1$. (c) $-7, 1, 0$. (d) $1, 0$.

- II.** Let A is a square matrix whose Eigen values are $0, 1, -1$ then A^{-1} is **01**
 (a) 0. (b) 1. (c) -1. (d) Does not exist.
- III.** Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ be a matrix then characteristic equation of matrix A is **01**
 (a) $\lambda^2 - \lambda + 1 = 0$.
 (b) $\lambda^2 - 2\lambda + 1 = 0$.
 (c) $\lambda^2 + 2\lambda + 1 = 0$.
 (d) $\lambda^2 + \lambda + 1 = 0$.
- IV.** If the vector $\vec{F}$ is irrotational then, $\text{Curl}(\vec{F})$ is **01**
 (a) 0. (b) 1. (c) -1. (d) We can't say anything.
- V.** The dot product of vectors $(1, 3)$ and $(3, 1)$ is **01**
 (a) 0. (b) 6. (c) -6. (d) -8.
- VI.** If $\vec{F}$ is conservative field then line integral is **01**
 (a) depend on starting and end point of the path.
 (b) depend on curve.
 (c) depend on region.
 (d) depend on path.
- VII.** Which one of the following theorem is transformation between surface integral and line integral? **01**
 (a) Gauss divergence theorem.
 (b) Green's theorem.
 (c) Stoke's theorem.
 (d) None of the options.
- Q-5 Attempt any Four.** **[16]**
- I.** Find the Eigen values and Eigen vectors of a matrix $\begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$. **04**
- II.** Find the characteristic equation of a matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence find **04**
 $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$.
- III.** Using Cayley-Hamilton theorem, Evaluate A^{-1} and A^2 , Where $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$. **04**
- IV.** If $\vec{V} = 3xz\hat{i} + 2xy\hat{j} - yz^2\hat{k}$ then find $\text{div}(\vec{V})$ at $(1, 0, 1)$. **04**
- V.** Evaluate line integral of $\vec{F} = xz\hat{i} + yz\hat{j} + z^2\hat{k}$, Where C is the curve **04**
 $x = t, y = t^2, z = t^3$ from the point $(0, 0, 0)$ to $(1, 1, 1)$.
- VI.** Evaluate $\int xy - ydx$ around the circle $x^2 + y^2 = 1$. **04**

Q-6(a) Attempt any Two. [06]

- I.** Prove that the vector $\vec{V} = (\sin y + z)\hat{i} + (x \cos y - z)\hat{j} + (x - y)\hat{k}$ is irrotational. **03**
- II.** Find gradient of a function $\phi(x, y, z) = 4xz^3 - 3x^2y^2z$ at the point $(2, -1, 2)$. **03**
- III.** Define Tangent vector and Normal vector. **03**

Q-6(b) Using Green's theorem Evaluate $\int_C \vec{F} \cdot d\vec{r}$ for $\vec{F}(x, y) = (3x^2 - 8y^2)\hat{i} + (4y - 6xy)\hat{j}$ [06]
where C is the boundary of the region bounded by $x = 0, y = 0, x + y = 1$.

OR

Q-6(b) Verify Stokes's theorem for $\vec{F} = x^2\hat{i} - xy\hat{j}$, where C is the square in the plane $z = 0$ [06]
and bounded by $x = 0, y = 0, x = a, y = a$.
